

Improved Projective Dynamics Global Using Snapshots-based Reduced Bases

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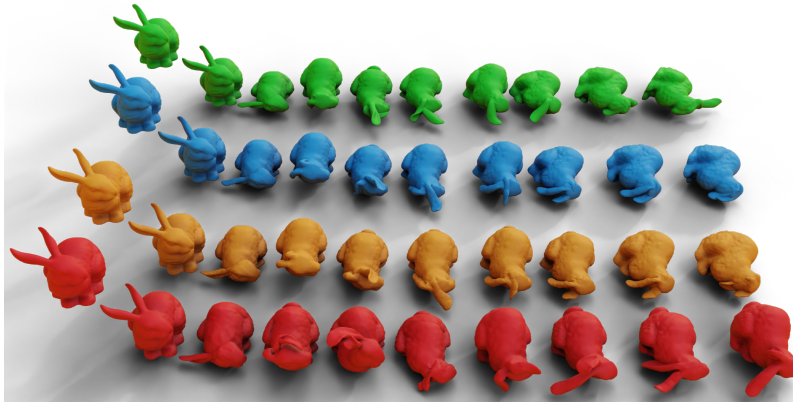


Figure 1: Visual comparison between full simulations (green) and reduced simulations by sparse PCA (blue) SPLOCS (orange) and LBS (red). A sequence of centered frames from a "falling bunny" simulations.

CCS CONCEPTS

• Computing methodologies → Real-time simulation.

KEYWORDS

model reduction, reduced subspaces, real-time simulation, physical simulation, projective dynamics

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1 INTRODUCTION

Physics-based simulations are widely used for real-time applications. Deformable objects pose significant computational challenges due to their complex nonlinear behavior and problem's high dimensionality. Many methods were developed to accelerate these simulations while maintaining accuracy and stability. These include explicit and implicit finite element methods [Snowden et al. 2017],

position-based dynamics [Bender et al. 2015, 2014], and projective dynamics [Bouaziz et al. 2014]. Each method has its own advantages and limitations in terms of speed, accuracy, and stability, providing different trade-offs for specific application requirements.

Hyper-reduced projective dynamics (HRPD) is a model reduction technique proposed by Brandt *et al.* [Brandt et al. 2018] that combines the advantages of projective dynamics and reduced bases methods. HRPD achieves significant speedups while maintaining high accuracy compared to other techniques. Another promising approach is based on snapshot methods, such as sparse proper orthogonal bases and sparse localized components, which identify the most significant deformation modes from a set of snapshots obtained from pre-computed simulations or experiments [Barbič and James 2005; Stanton et al. 2013].

2 APPROACH

Model reduction techniques for elastic objects start by eliminating the global rotational with respect to average positions. This is done in the pre-processing steps of sparse PCA, sparse localized components (SPLOCS) [Neumann et al. 2013], and co-rotational techniques [Huang et al. 2019]. We propose a novel approach that incorporates the rotational components and avoids the pre-alignment step, allowing us to construct bases components suitable for real-time interactive applications that account for both translation and rotations simultaneously. Our approach achieves faster global linear solve, thanks to the sparsity of the bases, while maintaining better accuracy and stability compared to HRPD [Brandt et al. 2018].

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3 METHOD AND IMPLEMENTATIONS

The motion trajectory of a mesh positions $q(t) \in \mathbb{R}^{n \times 3}$ under internal f_{ext} and external forces $f_{in}(q) = \sum_j W_j(q)$ along time t is governed by Newton's second law of motion

$$\ddot{q} = f_{in}(q) + f_{ext}. \quad (1)$$

The core idea of projective dynamics [Bouaziz et al. 2014] is to use a solver that alternates between the projection computation, through local solve, and quantifying the positions for the next frame by solving a global linear system for the new positions q_{k+1}

$$\underbrace{\left(\frac{M}{(\delta t)^2} + \sum_j \omega_j S_j^T S_j \right)}_A q_{k+1} = \underbrace{\frac{M}{(\delta t)^2} s_k(q_k)}_{b_k} + \underbrace{\sum_j S_j^T p_j(q_k)}_{f(t)} \quad (2)$$

at fixed constraints projections. $p_j(q)$ is the projection of the positions on the j^{th} constraint manifold C_j . The w_j are non-negative weights, S_j is a matrix that selects the elements influenced by C_j , and δt is the time step. The global linear solve (2) can be written as $Aq_{k+1} = Bu(t) + f(t)$, in which case A and $B = \frac{M}{(\delta t)^2}$ are called system matrices, $u(t) = s_k(q_k)$ is the time dependent input and $f(t)$ a nonlinear function that requires an update at each time step.

In vertex position reduction, we search a linear subspace $U \in \mathbb{R}^{n \times k}$, with $k \ll n$, such that $q \approx U\tilde{q}$ for some $\tilde{q} \in \mathbb{R}^{k \times 3}$. The linear solve then amounts to an approximate solution of a reduced system

$$U^T E U \tilde{q}(t) = U^T B u(t) + U^T f(U \tilde{q}(t)),$$

instead of the full one that is expensive to compute. Brandt *et al.* [Brandt et al. 2018] utilized linear blend skinning to construct reduced subspaces for positions and for constraints resulting in fast while physically inaccurate simulations. For analysis purpose we focus in this work on reducing vertex positions using different bases and compare to skinning position subspaces. Snapshot-base methods, such as proper orthogonal decomposition bases or principal components performed on collected full simulation frames [Alexa and Müller 2000], can be effective in reducing both dimension and approximation error of positions.

It is common practice in previous works to separate the geometric and animation-related aspects of the principal bases through a pre-processing step involving rigid alignment and removal of global rotation. However, we found that this leads to a degradation of the representation of rotational motion and decided to avoid rigid-alignment. We examined mass-weighted sparse PCA as well as SPLOCS components to compute U , see the Algorithm.

4 CONCLUSION

Our results in Figures 1 and 2 demonstrate improved physical behavior as well as a global linear solve that maintains a better time margin, by using sparse PCA and SPLOCS compared to LBS for projective dynamics real-time simulations. Future work will include exploring similar bases for constraint projection reduction.

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Algorithm 1: Weighted SPLOCS bases

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1 Input: non-rigid-aligned localized un-normalized PCA
  bases  $U = U_{pca}$ .
  for  $k$  in PCA components do
2    $id_k$  = vertex with largest deformation.
3   Compute support map for  $id_k$ 
4   Sparsify the extracted component  $c_k$ 
5 Optimize energy ADMM
6 Undo normalization for  $U$ 
7  $U_{splocs} \leftarrow M^{-\frac{1}{2}} U$ 

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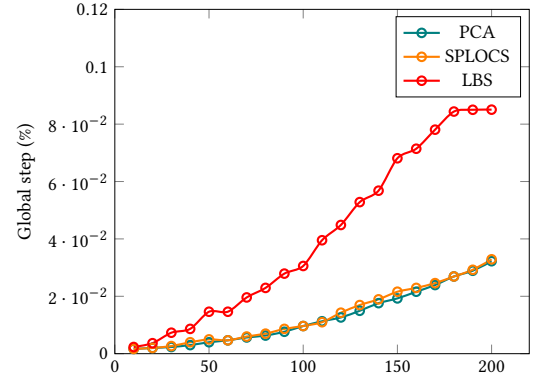


Figure 2: Relative global solve time for different bases.

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